A WEIGHT-SPLITTING TEST FOR R-UNIVERSALITY OF A SEMISIMPLE LIE ALGEBRA THAT SPLITS OVER A QUADRATIC EXTENSION

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Allowed standard prerequisites (editorial).

- The equivalence between a weight being star-invariant with vanishing beta and the irreducibility of the associated representation over the field. The author's proposition labelled HighWt, invoked inside the counted proof at source lines 226-231 with its labelled item HighWt-IrredIff. Converts the two hypotheses of the statement into irreducibility statements over R and over Q.
- The Frobenius-Schur type dichotomy for a Lie algebra that splits over a quadratic extension. Used inside the counted proof at source line 234 and attributed by the author to Corollary 3.2(2) of the cited paper Morris-RRepsQForms. Shows that the base change of the Q-representation to C is either irreducible or a sum of two irreducibles, which is the counting step of the proof.

Source reference. Original proposition 2.1: sources/algebra-fl02-a-cohomological-proof-that-real-re/pinned-original-source.tex, source lines 175–175. Navigation pointer

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only: the counted proof invokes this proposition twice for the equivalence between the star-invariance conditions and irreducibility; its statement is declared as a named standard prerequisite and is not reproduced.

Source reference. Original definition 2: sources/algebra-fl02-a-cohomological-proof-that-real-re/pinned-original-source.tex, source lines 188–188. Navigation pointer only: it is the labelled item of that proposition which the counted proof cites twice; the pointer keeps the author's own cross-reference resolvable.

Corollary 0.1. *Suppose* $\mathfrak{g}$ *is a semisimple Lie algebra over* $\mathbb{Q}$*, such that* $\mathfrak{g}$ *splits over a quadratic extension of* $\mathbb{Q}$*, and, for every dominant weight* λ *of* $\mathbf{T}$*:*

$$\begin{aligned}&\text{if } \lambda \text{ is invariant under the } *\text{-action of } \mathrm{Gal}(\mathbb{C}/\mathbb{R}), \text{ and } \beta_{\mathfrak{g},\mathbb{R}}(\lambda) = 0,\\ &\text{then } \lambda \text{ is also invariant under the } *\text{-action of } \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}), \text{ and } \beta_{\mathfrak{g},\mathbb{Q}}(\lambda) = 0.\end{aligned} \tag{0.1}$$

Then $\mathfrak{g}$ *is* $\mathbb{R}$*-universal.*

Proof. Since representations of $\mathfrak{g}$ are completely reducible, it suffices to show that every *irreducible* real representation of $\mathfrak{g}$ has a $\mathbb{Q}$-form. Specifically, we will show that ${}^{\mathbb{Q}}\rho_\lambda$ is a $\mathbb{Q}$-form of ${}^{\mathbb{R}}\rho_\lambda$, for every dominant weight λ.

We begin by showing that if ${}^{\mathbb{R}}\rho_\lambda \otimes_{\mathbb{R}} \mathbb{C}$ is irreducible, then ${}^{\mathbb{Q}}\rho_\lambda \otimes_{\mathbb{Q}} \mathbb{C}$ is irreducible. From Proposition 2.1(2) (with $F = \mathbb{R}$), we know that λ is invariant under the $*$-action of $\mathrm{Gal}(\mathbb{C}/\mathbb{R})$, and that $\beta_{\mathfrak{g},\mathbb{R}}(\lambda) = 0$. By assumption, this implies that λ is invariant under the $*$-action of $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$, and that $\beta_{\mathfrak{g},\mathbb{Q}}(\lambda) = 0$. Then, from Proposition 2.1(2) (with $F = \mathbb{Q}$), we conclude that ${}^{\mathbb{Q}}\rho_\lambda \otimes_{\mathbb{Q}} \mathbb{C}$ is irreducible, as desired.

Since $\mathfrak{g}_{\mathbb{Q}}$ splits over a quadratic extension, we know that ${}^{\mathbb{Q}}\rho_\lambda \otimes_{\mathbb{Q}} \mathbb{C}$ is either irreducible or the direct sum of two irreducibles [6, Corollary 3.2(2)]. Therefore, ${}^{\mathbb{Q}}\rho_\lambda \otimes_{\mathbb{Q}} \mathbb{C}$ and ${}^{\mathbb{R}}\rho_\lambda \otimes_{\mathbb{R}} \mathbb{C}$ have the same number of irreducible constituents. (Namely, either they are both irreducible, or they are both the direct sum of 2 irreducibles.) Since ${}^{\mathbb{R}}\rho_\lambda$ is a summand of ${}^{\mathbb{Q}}\rho_\lambda \otimes_{\mathbb{Q}} \mathbb{R}$, this implies that ${}^{\mathbb{R}}\rho_\lambda \cong {}^{\mathbb{Q}}\rho_\lambda \otimes_{\mathbb{Q}} \mathbb{R}$, so ${}^{\mathbb{Q}}\rho_\lambda$ is a $\mathbb{Q}$-form of ${}^{\mathbb{R}}\rho_\lambda$. □

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